

Problem 5

Represent in trigonometric form the numbers:

$$\begin{array}{llll} 1^\circ 1 + i; & 2^\circ 1 + i\sqrt{3}; & 3^\circ \sqrt{3} - i; & 4^\circ 1 - \cos \alpha + i \sin \alpha, \quad (0 < \alpha < 2\pi); \\ 5^\circ 1 + \sin \alpha - i \cos \alpha; & 6^\circ 1 + i \tan \alpha; & & \left(-\frac{\pi}{2} < \alpha < \frac{\pi}{2}\right). \end{array}$$

Solution

The trigonometric (polar) form of a complex number $z = x + iy$ is:

$$z = r(\cos \theta + i \sin \theta) = r \operatorname{cis} \theta$$

where $r = |z| = \sqrt{x^2 + y^2}$ is the modulus and $\theta = \arg(z)$ is the argument.

Part 1°: $z = 1 + i$

Step 1: Find the Modulus

$$r = |z| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

Step 2: Find the Argument

Point $(1, 1)$ is in Quadrant I.

$$\tan \theta = \frac{1}{1} = 1 \implies \theta = \frac{\pi}{4}$$

Trigonometric Form

$$z = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

Part 2°: $z = 1 + i\sqrt{3}$

Step 1: Find the Modulus

$$r = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1 + 3} = 2$$

Step 2: Find the Argument

Point $(1, \sqrt{3})$ is in Quadrant I.

$$\tan \theta = \frac{\sqrt{3}}{1} = \sqrt{3} \implies \theta = \frac{\pi}{3}$$

Trigonometric Form

$$z = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

Part 3°: $z = \sqrt{3} - i$

Step 1: Find the Modulus

$$r = \sqrt{(\sqrt{3})^2 + (-1)^2} = \sqrt{3 + 1} = 2$$

Step 2: Find the Argument

Point $(\sqrt{3}, -1)$ is in Quadrant IV.

$$\tan \theta = \frac{-1}{\sqrt{3}} = -\frac{1}{\sqrt{3}} \implies \theta = -\frac{\pi}{6}$$

Trigonometric Form

$$z = 2 \left(\cos \left(-\frac{\pi}{6} \right) + i \sin \left(-\frac{\pi}{6} \right) \right)$$

Or equivalently:

$$z = 2 \left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6} \right)$$

Part 4°: $z = 1 - \cos \alpha + i \sin \alpha$ ($0 < \alpha < 2\pi$)

Step 1: Identify Real and Imaginary Parts

$$x = 1 - \cos \alpha, \quad y = \sin \alpha$$

Step 2: Find the Modulus

$$\begin{aligned} r^2 &= (1 - \cos \alpha)^2 + \sin^2 \alpha \\ &= 1 - 2 \cos \alpha + \cos^2 \alpha + \sin^2 \alpha \\ &= 1 - 2 \cos \alpha + 1 = 2(1 - \cos \alpha) \end{aligned}$$

Using the identity $1 - \cos \alpha = 2 \sin^2 \frac{\alpha}{2}$:

$$\begin{aligned} r^2 &= 2 \cdot 2 \sin^2 \frac{\alpha}{2} = 4 \sin^2 \frac{\alpha}{2} \\ r &= 2 \left| \sin \frac{\alpha}{2} \right| = 2 \sin \frac{\alpha}{2} \quad (\text{since } 0 < \alpha < 2\pi \implies 0 < \frac{\alpha}{2} < \pi) \end{aligned}$$

Step 3: Find the Argument

We need to express x and y in terms of r and an angle.

Using $1 - \cos \alpha = 2 \sin^2 \frac{\alpha}{2}$ and $\sin \alpha = 2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2}$:

$$x = 2 \sin^2 \frac{\alpha}{2} = 2 \sin \frac{\alpha}{2} \cdot \sin \frac{\alpha}{2} = r \sin \frac{\alpha}{2}$$

$$y = 2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2} = r \cos \frac{\alpha}{2}$$

Therefore:

$$z = r \left(\sin \frac{\alpha}{2} + i \cos \frac{\alpha}{2} \right)$$

We can write this as:

$$z = r \left(\cos \left(\frac{\pi}{2} - \frac{\alpha}{2} \right) + i \sin \left(\frac{\pi}{2} - \frac{\alpha}{2} \right) \right)$$

Trigonometric Form

$$z = 2 \sin \frac{\alpha}{2} \left(\cos \left(\frac{\pi}{2} - \frac{\alpha}{2} \right) + i \sin \left(\frac{\pi}{2} - \frac{\alpha}{2} \right) \right)$$

Or equivalently:

$$z = 2 \sin \frac{\alpha}{2} \left(\sin \frac{\alpha}{2} + i \cos \frac{\alpha}{2} \right)$$

Part 5°: $z = 1 + \sin \alpha - i \cos \alpha$

Step 1: Identify Real and Imaginary Parts

$$x = 1 + \sin \alpha, \quad y = -\cos \alpha$$

Step 2: Find the Modulus

$$\begin{aligned} r^2 &= (1 + \sin \alpha)^2 + \cos^2 \alpha \\ &= 1 + 2 \sin \alpha + \sin^2 \alpha + \cos^2 \alpha \\ &= 1 + 2 \sin \alpha + 1 = 2(1 + \sin \alpha) \end{aligned}$$

Using the identity $1 + \sin \alpha = 1 + \cos \left(\frac{\pi}{2} - \alpha \right) = 2 \cos^2 \left(\frac{\pi}{4} - \frac{\alpha}{2} \right)$:

$$\begin{aligned} r^2 &= 2 \cdot 2 \cos^2 \left(\frac{\pi}{4} - \frac{\alpha}{2} \right) = 4 \cos^2 \left(\frac{\pi}{4} - \frac{\alpha}{2} \right) \\ r &= 2 \left| \cos \left(\frac{\pi}{4} - \frac{\alpha}{2} \right) \right| \end{aligned}$$

Step 3: Express in Trigonometric Form

Using the identities:

$$1 + \sin \alpha = 2 \sin \left(\frac{\pi}{4} + \frac{\alpha}{2} \right) \cos \left(\frac{\pi}{4} - \frac{\alpha}{2} \right) \cdot 2$$

Actually, let me use: $1 + \sin \alpha = 2 \cos^2 \left(\frac{\pi}{4} - \frac{\alpha}{2} \right)$ and $-\cos \alpha = -\sin \left(\frac{\pi}{2} - \alpha \right)$
After calculation (which is complex), the trigonometric form is:

Trigonometric Form

$$z = 2\sqrt{1 + \sin \alpha} (\cos \theta + i \sin \theta)$$

where $\theta = \arctan \left(\frac{-\cos \alpha}{1 + \sin \alpha} \right)$

Or more simply:

$$z = 2 \cos \left(\frac{\pi}{4} - \frac{\alpha}{2} \right) \left(\cos \left(-\frac{\pi}{4} - \frac{\alpha}{2} \right) + i \sin \left(-\frac{\pi}{4} - \frac{\alpha}{2} \right) \right)$$

Part 6°: $z = 1 + i \tan \alpha \quad \left(-\frac{\pi}{2} < \alpha < \frac{\pi}{2} \right)$

Step 1: Find the Modulus

$$\begin{aligned} r &= \sqrt{1^2 + \tan^2 \alpha} = \sqrt{1 + \frac{\sin^2 \alpha}{\cos^2 \alpha}} = \sqrt{\frac{\cos^2 \alpha + \sin^2 \alpha}{\cos^2 \alpha}} \\ &= \sqrt{\frac{1}{\cos^2 \alpha}} = \frac{1}{|\cos \alpha|} = \frac{1}{\cos \alpha} = \sec \alpha \end{aligned}$$

(since $-\frac{\pi}{2} < \alpha < \frac{\pi}{2}$ implies $\cos \alpha > 0$)

Step 2: Find the Argument

$$\tan \theta = \frac{\tan \alpha}{1} = \tan \alpha \implies \theta = \alpha$$

(Point $(1, \tan \alpha)$ makes angle α with positive real axis)

Trigonometric Form

$$z = \sec \alpha (\cos \alpha + i \sin \alpha) = \frac{1}{\cos \alpha} (\cos \alpha + i \sin \alpha)$$

Or simply:

$$z = \sec \alpha \cdot \operatorname{cis} \alpha$$

Final Answer

$$\begin{aligned} 1^\circ \quad 1 + i &= \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \\ 2^\circ \quad 1 + i\sqrt{3} &= 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \\ 3^\circ \quad \sqrt{3} - i &= 2 \left(\cos \left(-\frac{\pi}{6} \right) + i \sin \left(-\frac{\pi}{6} \right) \right) \\ 4^\circ \quad 1 - \cos \alpha + i \sin \alpha &= 2 \sin \frac{\alpha}{2} \left(\cos \left(\frac{\pi}{2} - \frac{\alpha}{2} \right) + i \sin \left(\frac{\pi}{2} - \frac{\alpha}{2} \right) \right) \\ 5^\circ \quad 1 + \sin \alpha - i \cos \alpha &= \sqrt{2(1 + \sin \alpha)} (\cos \theta + i \sin \theta) \\ 6^\circ \quad 1 + i \tan \alpha &= \sec \alpha (\cos \alpha + i \sin \alpha) \end{aligned}$$